INCH-POUND

MIL-T-25905C(USAF)
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SUPERSEDING
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MILITARY SPECIFICATION

# TRANSMITTER, RATE OF FLOW, TRU-7/A
# FUEL, 0-15, 000 PPH

## 1. SCOPE

1.1 Scope. This specification covers one type of remote indicating, fuel, rate of flow transmitter, designated TRU-7/A.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | Box, Wood, Cleated Plywood |
| PPP-B-636 | Box, Shipping, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-J-5624 | Turbine Fuel Aviation, Grades JP4 And JP5 |
| MIL-I-7566 | Indicator System, Fuel Flowmeter, General Specification For |
| MIL-P-8422 | Power Supply, 28V DC Input, 15V 4 Cycle 3 Phase Output, Type MC-1 |
| MIL-I-25906 | Indicator, Rate Of Flow, EFU-3/A, Fuel, 0-15,000 PPH |

Beneficial comments (recommendations, additions, deletions) any any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TIMLA, Tinker AFB OK 73145-5990 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking of US Military Property |
| MIL-STD-810C | Environmental Test Methods |
| MIL-STD-2073/1A | DOD Materials Procedures For Developing And Application Of Packaging Requirements |
| MS33649 | Boss, Fluid Connection-Internal-Straight Thread |
| MS33678 | Connector, Receptable, Electric Integral, Mounting |

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Order Desk, Bldg 4D, 700 Robbins Ave, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American National Standards Institute (ANSI)

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be address to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 General specification. The requirements of MIL-I-7566 apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction.



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3.3.1 General. The transmitter shall be designed and calibrated to operate satisfactorily with 0-15,000 pph indicator in accordance with MIL-I-25906, and power supply in accordance with MIL-P-8422.

3.3.1.1 Remote indication. The transmitter remote indication system shall operate on the second harmonic system in which the moving element consists of a permanent magnet, and the toroidally wound stator is energized with 115V 400 cycle single phase alternating current.

3.3.2 Range. The transmitter shall measure fuel flow rated from 0 to 15,000 pph with a suppressed zero stop in the flow range of 100 to 275 pph. This stop shall form a part of the permanent mechanical stop which shall be provided within the transmitter to prevent damage due to excessive flow rates.

3.3.3 Operating pressure. The high pressure transmitter shall operate under a pressure of 600 psi and withstand a pressure of 1600 psi without damage.

3.3.4 Pressure drop. The transmitter, by virtue of its design, shall not have appreciably more pressure drop under any fuel flow conditions when in operative, than when it is operating normally.

3.3.5 Temperature range. The transmitter shall be designed to withstand an ambient operating temperature range of -55°C to +121°C with fuel flowing through the transmitter, and exposure to -65°C to +190°C with no fuel flowing through the transmitter.

3.3.6 Electrical receptacles. The transmitter's electrical receptacles shall be in accordance with MS33678C14S-7P for the operating power source, and MS33678C14S-2P for connection to the flow rate indicator in accordance with figure 1, except that the receptacles shall be designed for hermetic sealing.

3.3.7 Power consumption. The power consumption of the pickoff coil shall be a maximum of 3.0 watts.

3.3.8 Dimensions. The outline dimensions of the transmitter shall conform to figure 2.

3.3.9 Weight. The weight of the complete fuel flowmeter transmitter shall not exceed 6.5 pounds.

3.3.10 Mounting. Mounting bosses and spacing thereof shall be as specified in figure 2.

3.3.11 Flow direction. The direction of fuel flow through the transmitter shall be permanently and legibly marked by arrow as shown on figure 2 on both front and back.

3.3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.4 Recovered materials. Recovered materials shall be used to the maximum extent possible without jeopardizing the end use of the item.



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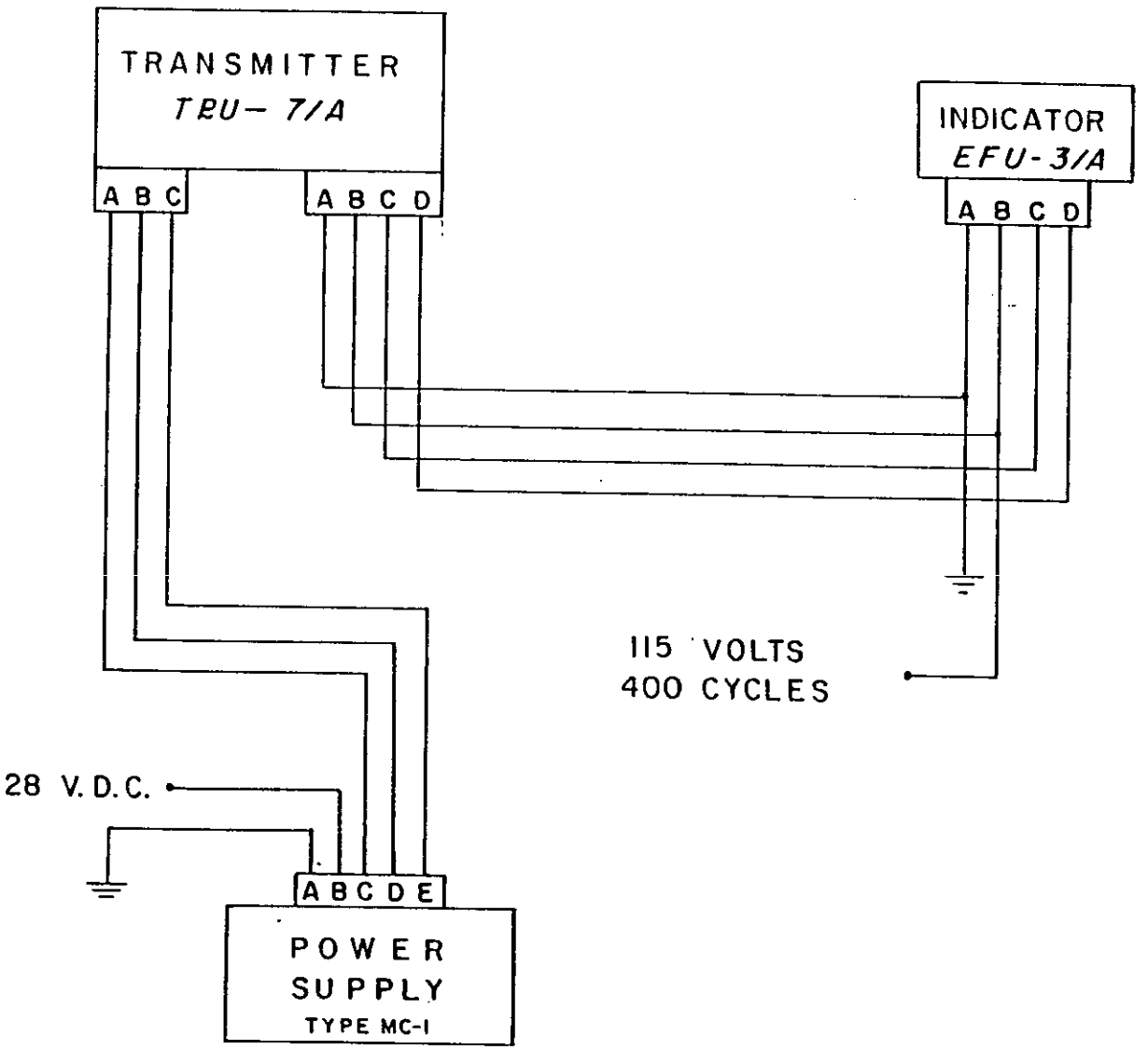



FIGURE 1. External wiring.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.1.1 General. Samplings inspection and testing of the transmitters shall be as specified in MIL-I-7566, and as specified herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in section 4.

4.3.1 Tolerances. Applicable tolerances shall be as specified in section 3, and as specified in tables I, II, and III.

4.3.2 Test position. Unless otherwise specified the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.3.3 Test fuel. The transmitter shall be tested in accordance with MIL-F-5624 using aviation fuel having a specific gravity of 0.785 $\pm$0.005 at 15.6°C. A commercial equivalent of higher flash point may be used for room temperature and high temperature tests.

4.3.4 Fuel temperature. The temperature of the test fuel during high temperature test runs shall be 38 $\pm$ 2°C. The fuel shall be pumped through the transmitter at the rate of 1000 pph for the last 30 minutes prior to scale and friction error calibration tests.

4.4 First article inspection.



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TABLE I

Fuel Flowmeter Transmitter.

| Flow PPH | Scale Error Tolerance PPH± | |
|---|---|---|
| | Substitute Fuel and Room Temperature | High Temperature | Low Temperature |
| 500 | 20 | 25 | 30 |
| 1,000 | 25 | 30 | 40 |
| 1,500 | 30 | 40 | 50 |
| 2,000 | 30 | 40 | 60 |
| 2,500 | 35 | 45 | 70 |
| 3,000 | 50 | 45 | 80 |
| 4,000 | 200 | 250 | 240 |
| 6,000 | 220 | 275 | 280 |
| 9,000 | 250 | 315 | 340 |
| 12,000 | 280 | 350 | 400 |
| 15,000 | 310 | 390 | 460 |

TABLE II

Fuel Flowmeter Transmitter.

| Test | Tolerance PPH | |
|---|---|---|
| | 500 - 3000 Scale | 3000 - 15,000 Scale |
| Voltage and frequency variation | 10 | 60 |
| Position error | 20 | 120 |
| Friction | | |
| Room temperature | 10 | 60 |
| High temperature | 10 | 60 |
| Low temperature | 10 | 60 |
| Vibration | | |
| Pointer oscillation | 30 | 180 |
| Pointer variation | 20 | 160 |
| Power consumption 2.2 maximum | | |
| Up-scale - down scale reading variation | 25 | 80 |

TABLE III

Fuel Flowmeter Transmitter.

| Flow Rate PPH | Maximum Allowable Pressure Drop PSI |
|---|---|
| 6,000 | 0.21 |
| 9,000 | 0.46 |
| 12,000 | 0.81 |
| 15,000 | 1.3 |



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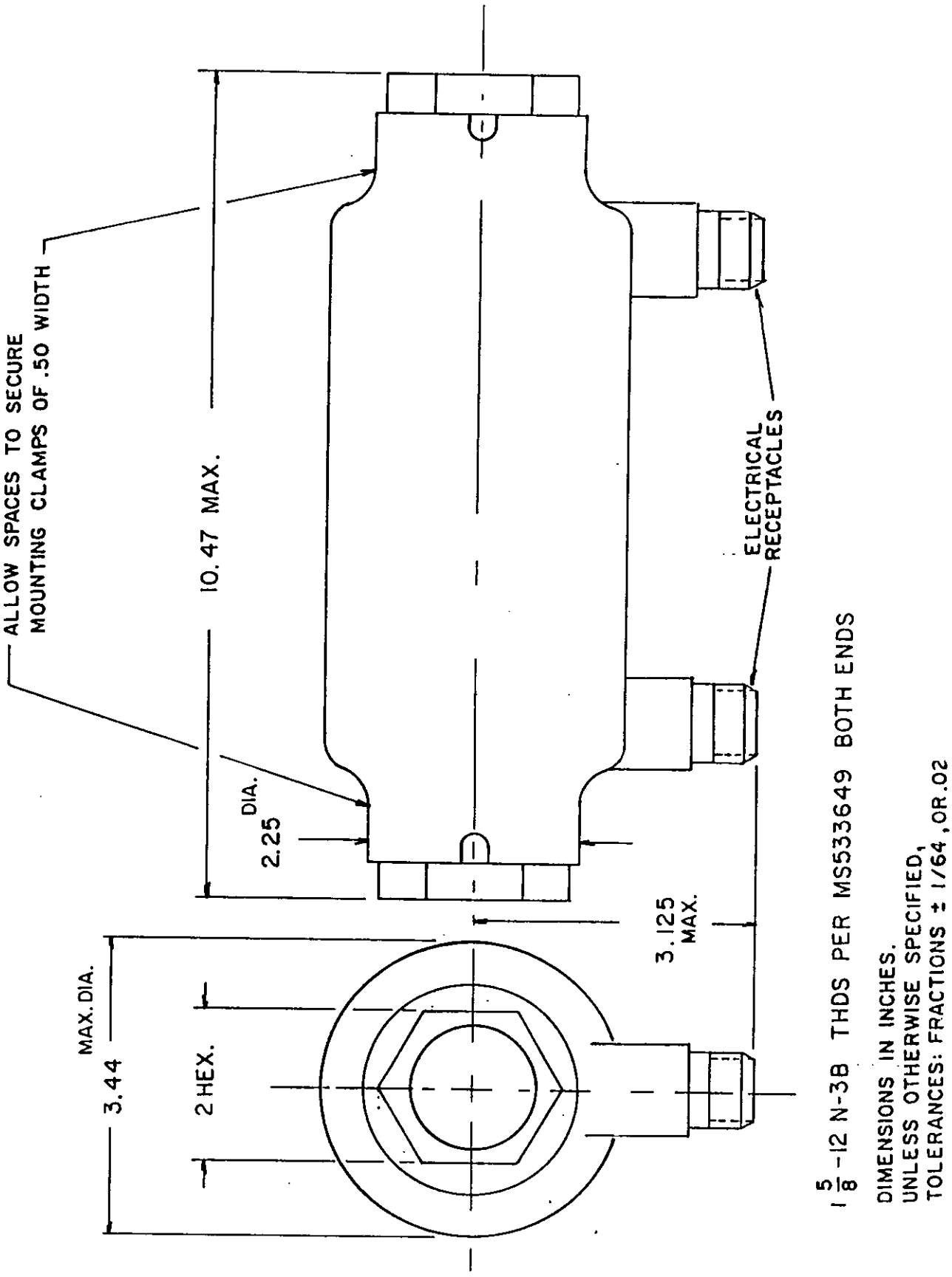



FIGURE 2. Transmitter.

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4.4.1 First article samples. The contractor shall subject three transmitters to tests specified in 4.4.4.

4.4.2 Sample disposition. The contractor shall, submit the tested samples to the responsible activity who will use them for:

a. Review of the mechanical construction of the product.

b. Performance of any tests included in first article inspection.

4.4.3 First article inspection testing. The first article tests shall consist of all the tests specified in MIL-I-7566 and herein.

4.5 Quality conformance inspection.

4.5.1 Group A inspection.

4.5.1.1 Pressure drop. The pressure drops of the transmitter shall be determined for the rates of flow specified and shall not exceed the tolerances specified in table III.

4.5.1.2 Transmitter vibration. The pointer oscillation and variation test errors shall be determined when the transmitter is subjected to the frequency range at vibratory accelerations or double amplitudes shown below:

| FREQUENCY | AMPLITUDE |
|---|---|
| 5-100 cps | 0.010 inch double amplitude |
| 100-500 cps | 5g acceleration |

4.5.2 Group B inspection.

4.5.2.1 Altitude test. Not required.

4.5.2.2 Low temperature exposure. The transmitter shall then be subjected to a temperature of -65° ±2°C for a period of not less than 24 hours, and then operated at the temperature over the scale range to determine that no component parts have been damaged. The temperature shall then be raised to -54° ±2°C for a period of not less than 24 hours. The scale errors and friction shall not exceed the low temperature tolerances as shown in tables I and II. The transmitter shall be subjected to the room temperature scale error test not less than one hour following the cold exposure test. The room temperature tolerances as specified in tables I and II shall apply.

4.5.2.3 High temperature exposure. The transmitter shall be subjected to an ambient temperature of 121° ±2°C for a period of four hours with fuel in the instrument, but not flowing. During the last 30 minutes of this period, fuel at 38° ±2°C shall be pumped through the transmitter at a rate of 1,000 pph. The transmitter shall then be tested for scale errors and friction under the same conditions. The scale errors and friction shall not exceed the tolerances specified in tables I and II.

4.5.2.4 Endurance. The scale errors after endurance test shall not exceed the tolerances specified in table I for the room temperature scale error test by more than 20 pph in the lower scale range, and 90 pph in the upper range. Friction errors shall be within the tolerances specified in table II.



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4.5.2.4.1 Endurance duration. In the event the elapsed time of the endurance test is less than 1,200 hours when the specified number of flow applications has been completed, then the system shall be operated at 80 ± 5 percent of the range of the system until a total of 1,200 hours of operation has been completed.

4.5.2.5 Contaminated fuel operation. Test fuel containing dirt to the extent of 80 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for one hour. A scale and friction error test shall then be made with the contaminated fuel and operation shall be satisfactory and within the room temperature tolerances. This foreign matter shall be considered to consist of not less than 68 percent Si02 and shall have a particle size analysis as shown in table IV.

TABLE IV. Particle size.

| Foreign Particle Size, Microns | Percent of Total Weight |
|---|---|
| 0-5 | 30±2 |
| 5-10 | 18±3 |
| 10-20 | 16±3 |
| 20-40 | 18±3 |
| Over 40 | 9±3 |
| Through a 200-mesh screen | 100 Percent |

4.5.2.6 Response time. The fuel flow shall be changed from 0 to 6,000 pph by means of a rapid action cut-off valve in the fuel line. The time for the indication of a master indicator connected to the transmitter to come within one percent of its steady state position shall be within five seconds.

4.5.2.7 Shock test. The test shall be in accordance with method 516.2, procedure 1 of MIL-STD-801C. The transmitter shall be subjected to impact shocks of 15g, each shock impulse having a time duration of 7 ± 2 milliseconds. Three shocks shall be applied in each of the following directions:

a. Along the longitudinal axis.
b. At 45' to the longitudinal axis in one direction.

4.6 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116, to insure compliance with section 5 of this specifications.

## 5. PACKAGING

5.1 Preservation. Preservation shall be Level A, C, or Industrial IAW MIL-STD-2073/1A, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process C-1 of MIL-P-116.



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5.1.1.3 Preservative application. Preservative shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1C-1 of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 fiberboard container, weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the Transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of the transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be Level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare, and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, Class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The TRU-7/A fuel flowmeter transmitter covered by this specification is intended to indicate the rate of flow of fuel in pounds per hour being consumed by an aircraft jet engine when used in conjunction with a suitable indicator and power supply.

6.2 Acquisition requirements. Acquisition documents must specify the following:



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a. Title, number and date of this specification.
b. Level of packaging and packing required.
c. Number of samples and point of testing.
d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 First article. A first article is required for inspection and approval (see 3.1 and 4.4).

6.4 Subject term (key word) listing.

Style, second harmonic

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Preparing Activity:
AIR FORCE -71

Project No:
6620-F480



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
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3. DOCUMENT TITLE
MIL-T-25905C(USAF)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial (2) AUTOVON |
|---|---|
| c. ADDRESS (Include Zip Code) | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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